

Formal verification - Tutorial 10

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Rational series

Let Σ be a finite alphabet. A *series* is a mapping $f : \Sigma^* \rightarrow \mathbb{Q}$. The *support* of a series f is the language $\text{supp}(f) = \{w \in \Sigma^* \mid f(w) \neq 0\}$. Recall that the set of rational series coincides with the set of series recognised by weighted finite automata.

Exercise 1. Show an example of a rational series with a non-regular support.

Solution. Consider the series mapping a word $w \in \Sigma^*$ to the difference of the number of occurrences of a and b in w . The support of this series is the language of words with an unequal number of a 's and b 's, which is well-known not to be regular. The series is rational, in fact a weighted automaton of dimension two suffices to recognise it. $\square$

The *left shift* (or *left quotient*, or *left derivative*) of a series f by a word $w \in \Sigma^*$ is the series $w^{-1}f$ defined by $(w^{-1}f)(u) = f(wu)$ for all $u \in \Sigma^*$. The *rank* of a series f is the dimension of the vector space generated by the set of all left shifts of f , i.e., the set $\{w^{-1}f \mid w \in \Sigma^*\}$. Recall that the rank of a series is finite if, and only if, the series is rational.

Exercise 2. Assume that f, g are rational series with rank k , resp., ℓ .

1. Show that the rank of $f + g$ is at most $k + \ell$.
2. The Hadamard product of f and g is the series $f \odot g$ defined by $(f \odot g)(w) = f(w)g(w)$ for all $w \in \Sigma^*$. Show that the rank of $f \odot g$ is at most $k \cdot \ell$.

A weighted automaton is *observable* at an initial condition $u \in \mathbb{Q}^{1 \times k}$ if, for every initial condition $u' \in \mathbb{Q}^{1 \times k}$, $\llbracket A_u \rrbracket = \llbracket A_{u'} \rrbracket$ implies $u = u'$. A weighted automaton is *observable* if it is observable at every initial condition.

Exercise 3. Show that the observability problem for weighted automata is decidable in polynomial time.

A *parametric weighted automaton* is a tuple

$$A = (\Sigma, k, \ell, u, M, v)$$

where

- Σ is a finite input alphabet,

- $k \in \mathbb{N}$ is the dimension of the state
- $\ell \in \mathbb{N}$ is the number of parameters,
- $u \in \mathbb{Q}[p_1, \dots, p_\ell]^{1 \times k}$ is the parametric initial vector,
- $M : \Sigma \rightarrow \mathbb{Q}[p_1, \dots, p_\ell]^{k \times k}$ is the *transition function* mapping each letter to a parametric transition matrix, and
- $v \in \mathbb{Q}[p_1, \dots, p_\ell]^{k \times 1}$ is the parametric final vector.

For every given valuation for the parameters $\nu \in \mathbb{Q}^\ell$ we obtain a concrete weighted automaton $A(\nu)$ in the expected way. This concrete automaton will then recognise the rational series $\llbracket A(\nu) \rrbracket : \Sigma^* \rightarrow \mathbb{Q}$.

Exercise 4. *Show that the set of all parameter valuations $\nu \in \mathbb{C}^\ell$ s.t. $A(\nu)$ recognises the zero series is an algebraic variety. Show that polynomial generators of degree polynomial in k suffice to define this variety.*

The *parameter identifiability problem* for a parametric weighted automaton A amounts to deciding whether, for all parameter valuations $\nu, \nu' \in \mathbb{C}^\ell$, $\llbracket A(\nu) \rrbracket = \llbracket A(\nu') \rrbracket$ implies $\nu = \nu'$.

Exercise 5. *Show that the parameter identifiability problem is decidable for parametric weighted automata.*